

잡음 환경에서의 신호 품질 기반 동적 모델 선택을 이용한 베어링 결함 진단

SNR-Guided Dynamic Model Selection for
Bearing Fault Diagnosis under Noisy Conditions

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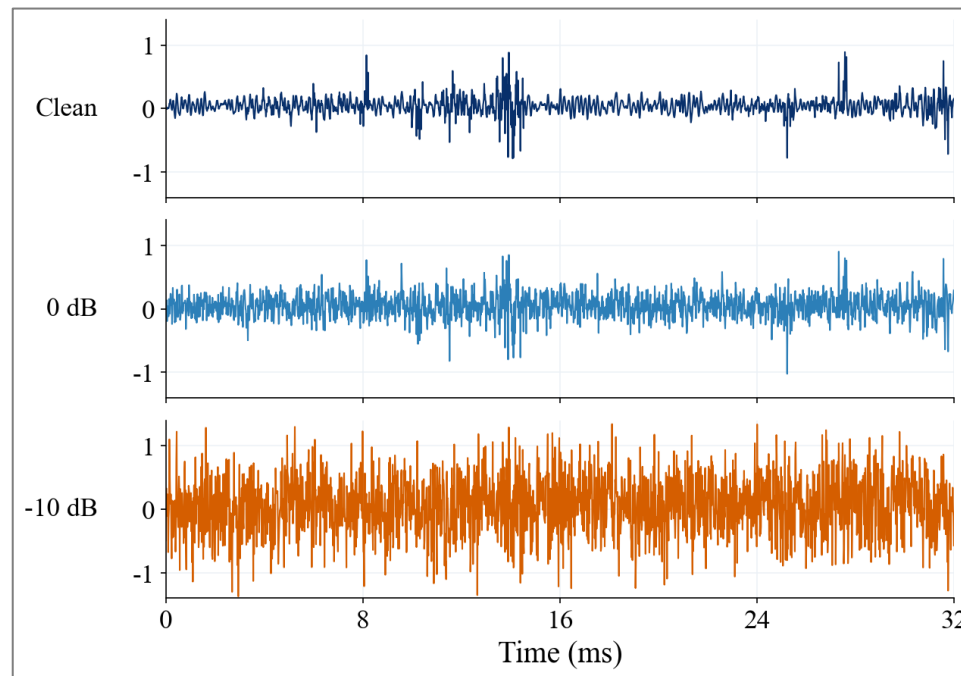
2026.06.26

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Bearing fault diagnosis from vibration signals

- Industrial vibration signals are often noisy.
- Noise makes fault patterns harder to distinguish.
- Continuous monitoring requires efficient inference.

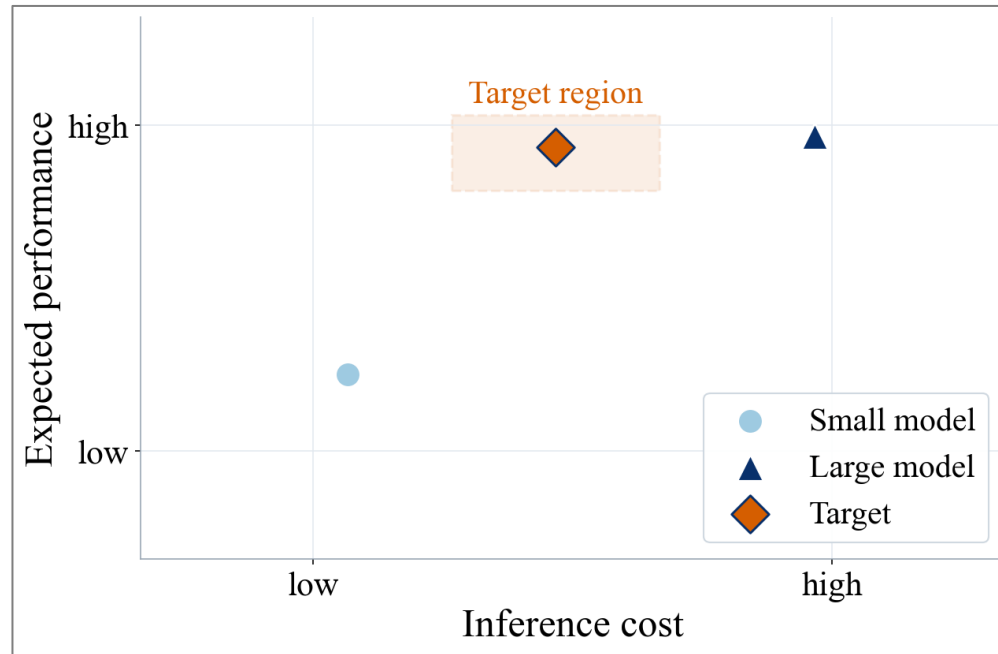


AWGN increasingly masks fault patterns

i. *Lessmeier et al., 2016, Condition Monitoring of Bearing Damage in Electromechanical Drive Systems by Using Motor Current Signals of Electric Motors: A Benchmark Data Set for Data-Driven Classification. PHM Society European Conference, 3(1).*

A Capacity–Cost Dilemma

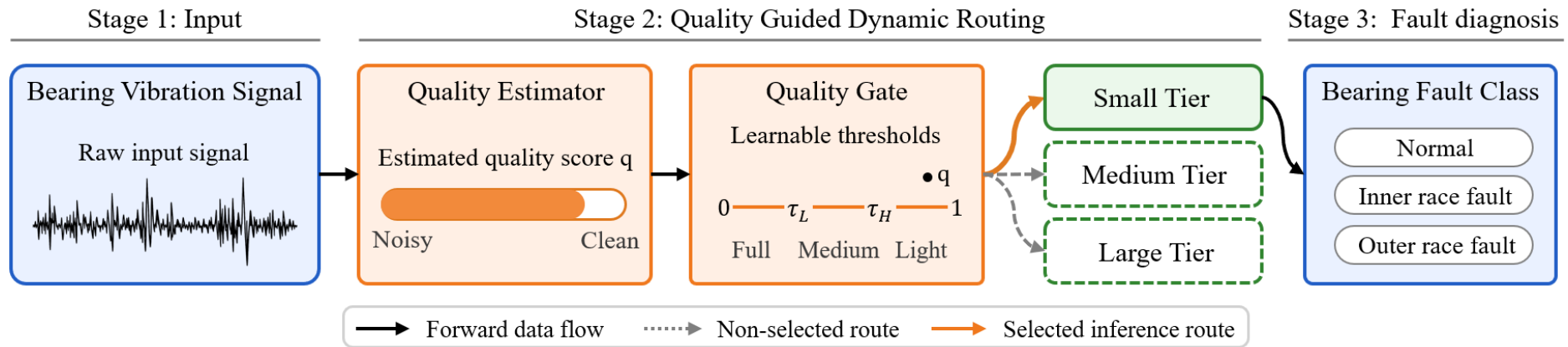
- Small models save cost but degrade under noise.
- Large models improve robustness but increase cost.
- Goal: reduce cost with minimal performance degradation.



Conceptual cost–performance trade-off

Quality-Guided Dynamic Routing

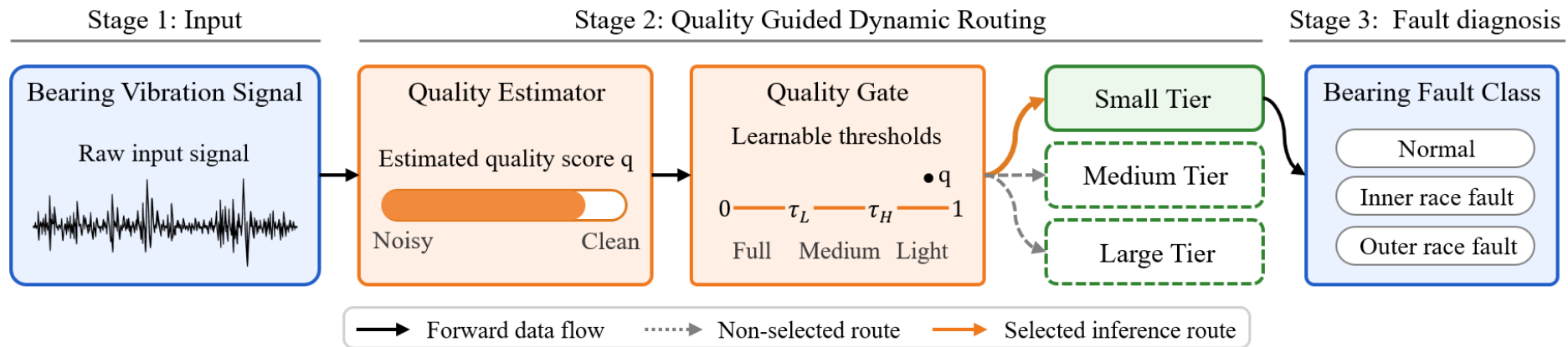
- Estimate a quality score q for each window.
- Select one 1D-CNN tier (small / medium / large).
- Run only the selected classifier.



Method

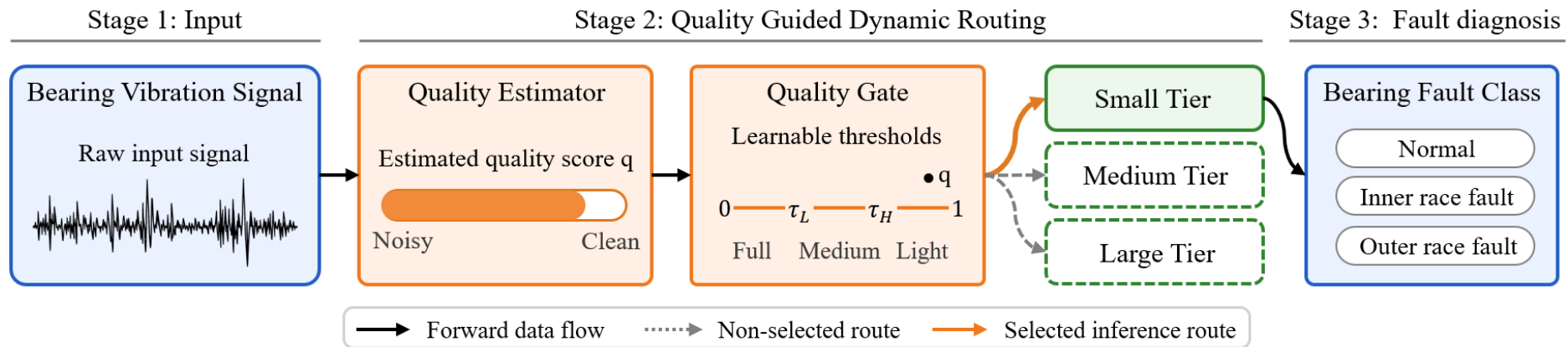
Quality Estimator

- Tiny 1D-CNN quality estimator (0.13K).
- Supervised by injected SNR, without fault labels.
- Outputs $q \in [0, 1]$, where higher q means a cleaner signal.
- Used only for routing.



Quality Gate

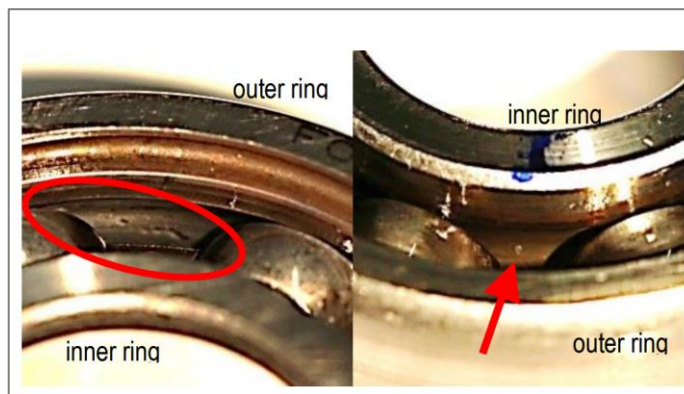
- Two thresholds calibrated on validation.
- High-quality inputs use the small tier.
- Low-quality inputs use the large tier.
- One model runs per window.



Experimental Setup

Dataset & Evaluation Protocol

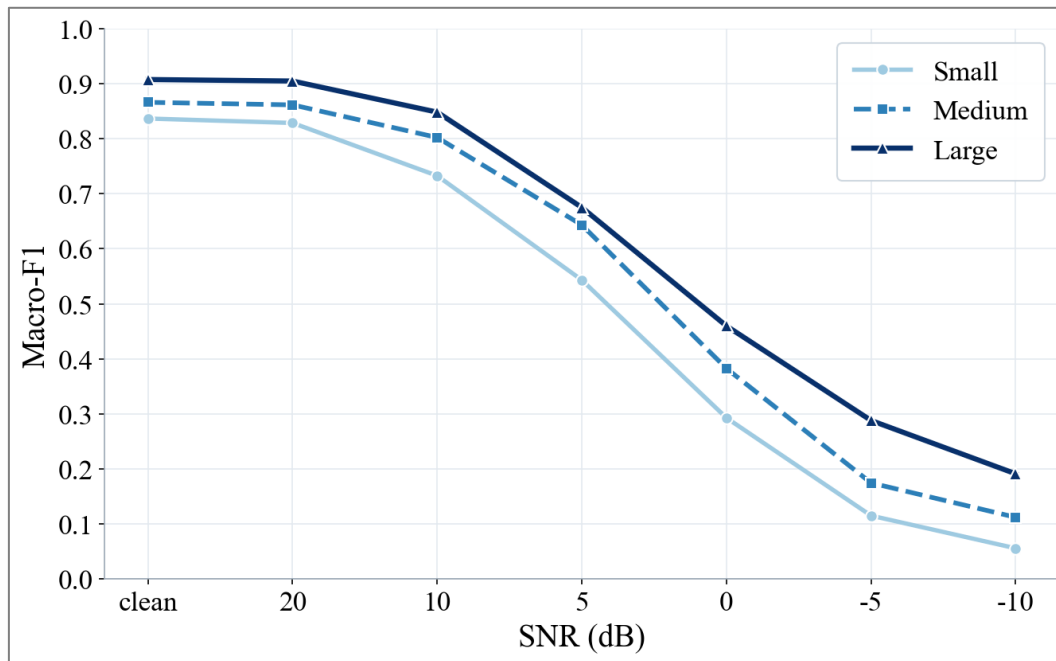
Property	Value
Dataset	Paderborn University (PU) bearing
Classes	3 (Normal / Inner race / Outer race)
Training	Clean signals only
Test	AWGN, per-window mixed SNR (clean to -10 dB)
Model tiers	Small 133K / Medium 331K / Large 524K (1D-CNN)
Evaluation metric	Macro-F1
Cost	Average active params



- i. *Lessmeier et al., 2016, Condition Monitoring of Bearing Damage in Electromechanical Drive Systems by Using Motor Current Signals of Electric Motors: A Benchmark Data Set for Data-Driven Classification. PHM Society European Conference, 3(1).*

Capacity improves noise robustness

- SNR is signal power over noise power.
- Macro-F1 is the average F1-score across classes.
- Larger models stay robust as SNR drops.

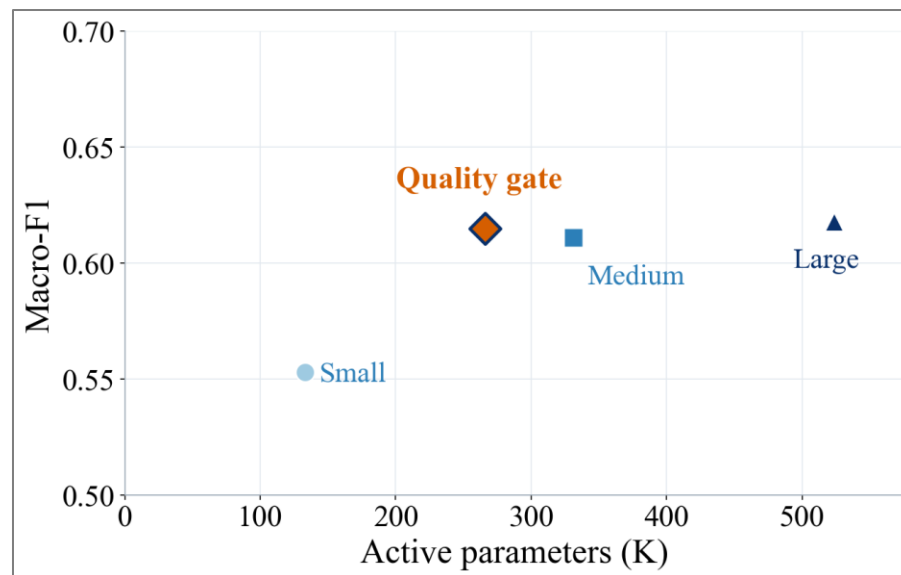


Per-SNR Macro-F1

Accuracy–Cost Trade-off

- Comparable F1 to the large model, within -0.3 pp.
- 49% fewer active params than large.
- 20% fewer active params than medium.

Method	Macro-F1	Active params	Param. ratio (Large=1)
Small	0.553	133K	0.25
Medium	0.611	331K	0.63
Large	0.618	524K	1.00
Quality gate	0.615	266K	0.51



Routing distribution

- Clean inputs route mostly to Small.
- Noisy inputs route mostly to Large.
- The gate scales capacity with noise.

SNR	Small	Medium	Large
clean	99.7	0.3	0.0
20 dB	99.3	0.6	0.1
10 dB	48.9	20.4	30.7
5 dB	40.0	2.8	57.2
0 dB	40.0	0.0	60.0
-5 dB	11.3	28.7	60.0
-10 dB	0.0	40.0	60.0

Routing share per SNR (%)

Cost-efficient routing at near-large F1.

- Dynamic model selection based on estimated signal quality.
- AWGN-injected windows provide known quality targets.
- Comparable Macro-F1 with 49% fewer average active parameters.

Future work

- Automatically optimize tier count and model sizes.



KETI
embedded

Thank you

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